a01_2

March 19, 2026

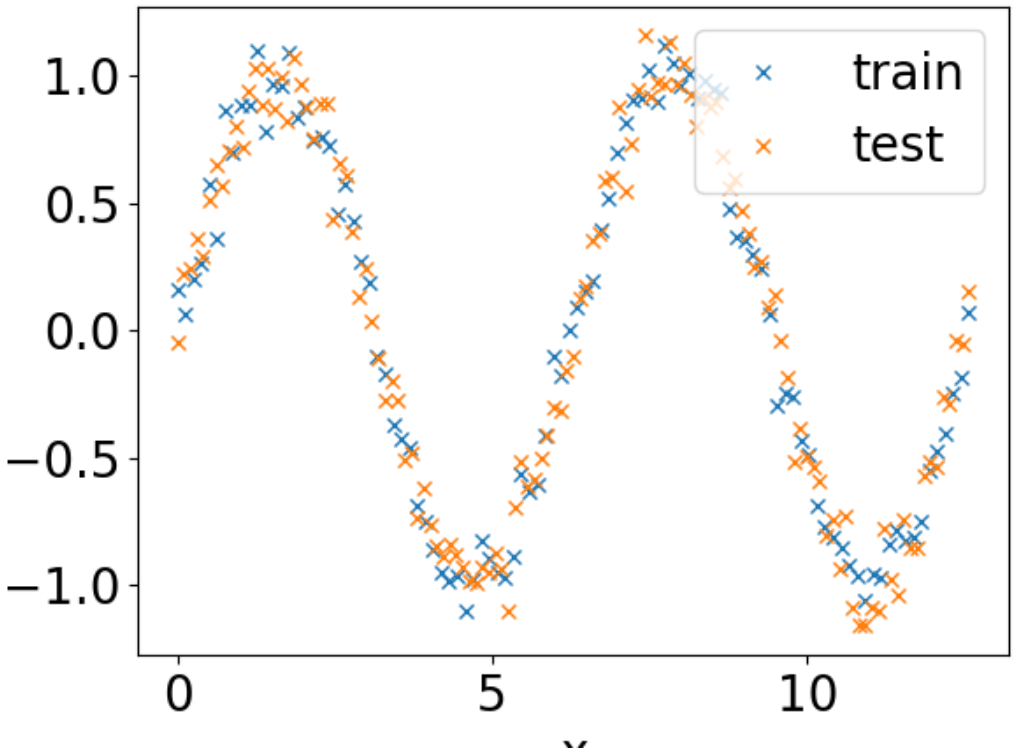
[15]:
```
import matplotlib.pyplot as plt
import torch
import torch.nn.functional as F

# import helper functions
import sys, os

sys.path.append(os.getcwd())
%load_ext autoreload
%autoreload 2
from a01_helper import *
from a01_functions import train1, MLP  # MLP is implicitly required for `torch.
↪load`
```

```
The autoreload extension is already loaded. To reload it, use:
  %reload_ext autoreload
```

1 2 Multi-Layer Feed-Forward Neural Networks

1.1 2a Conjecture how an FNN fit will look like

[16]:
```
# here is the one-dimensional dataset that we will use
nextplot()
plot1(X1, y1, label="train")
plot1(X1test, y1test, label="test")
plt.legend(loc="upper right")

# Use this function to save your plot.
saveplot("a01_2_data.pdf")
```



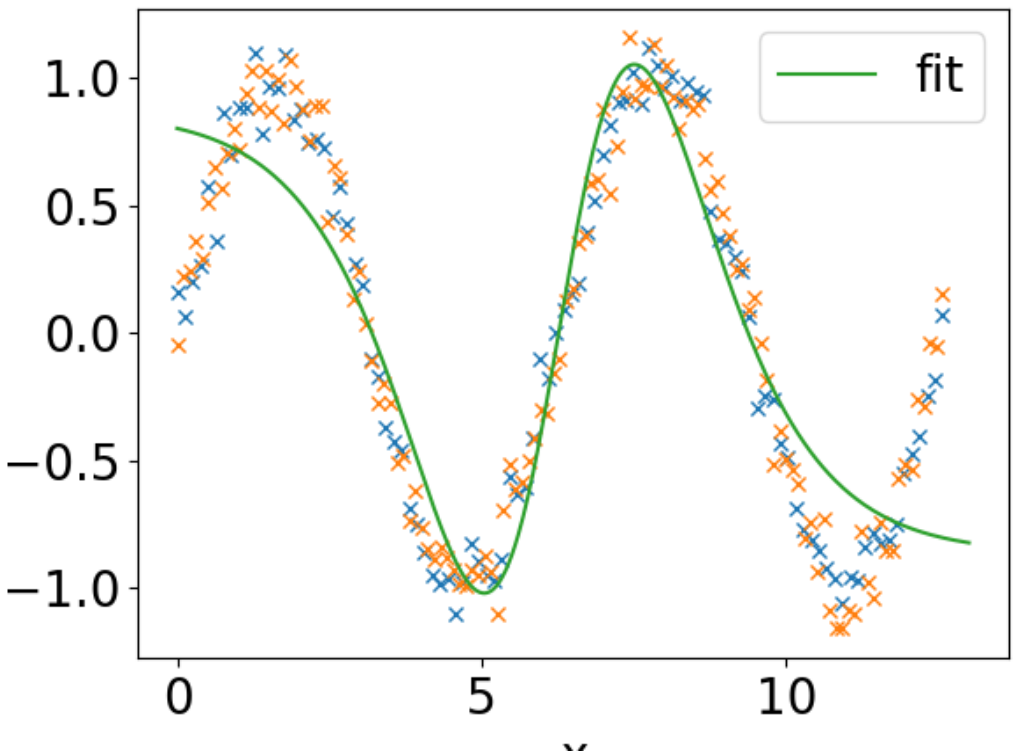
1.2 2b Train with 2 hidden units

[17]:
```
# Let's fit the model with one hidden layer consisting of 2 units.
model = train1([2], nreps=1)
print("Training error:", F.mse_loss(y1, model(X1)).item())
print("Test error    :", F.mse_loss(y1test, model(X1test)).item())
```

```
X1 shape: torch.Size([100, 1])
Repetition  0:          Current function value: 0.079573
         Iterations: 387
         Function evaluations: 517
         Gradient evaluations: 507
best_cost=0.080
Training error: 0.07957357168197632
Test error    : 0.08671440184116364
```

```
/home/tu2/anaconda3/envs/deep-learning/lib/python3.13/site-
packages/scipy/optimize/_minimize.py:779: OptimizeWarning: Desired error not
necessarily achieved due to precision loss.
  res = _minimize_bfgs(fun, x0, args, jac, callback, **options)
```

[18]:
```
# plot the data and the fit
nextplot()
plot1(X1, y1, label="train")
plot1(X1test, y1test, label="test")
plot1fit(torch.linspace(0, 13, 500).unsqueeze(1), model)

saveplot("a01_2_fit-2-neurons.pdf")
```



[19]:
```
# The weight matrices and bias vectors can be read out as follows. If you want,␣
↪use
# these parameters to compute the output of the network (on X1) directly and␣
↪compare to
# vmap(model)(X1).
for par, value in model.state_dict().items():
    print(f"{par:<15}= {value}")
```

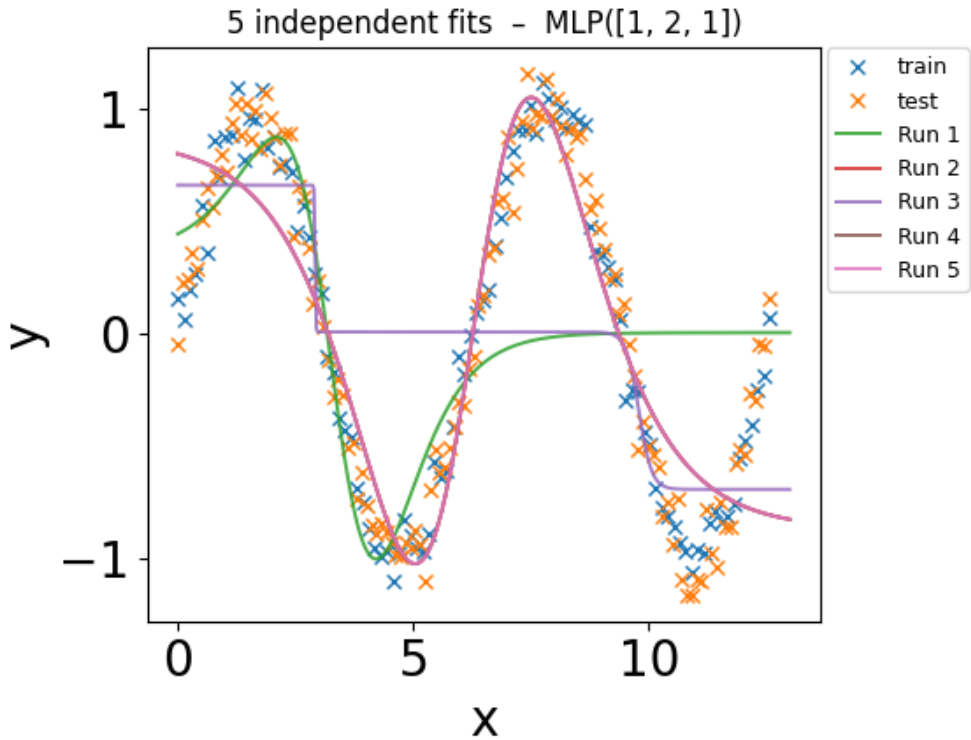
```
0_weight       = tensor([[-1.0559,  1.0695]])
0_bias         = tensor([ 6.6317, -6.7171])
1_weight       = tensor([[547.1893],
        [545.4597]])
1_bias         = tensor([-546.3256])
```

[20]:
```
# now repeat this multiple times
# TODO: YOUR CODE HERE
nextplot()

plot1(X1, y1, label="train")
plot1(X1test, y1test, label="test")

X_plot = torch.linspace(0, 13, 500).unsqueeze(1)
colors = plt.cm.tab10.colors   # 10 distinct colours from matplotlib's default␣
↪cycle
for i in range(5):
    print(f"\n--- Run {i+1} ---")
    temp_model = train1([2], nreps=1)
    lines = plt.plot(
        X_plot.numpy(),
        temp_model(X_plot).detach().numpy(),
        label=f"Run {i+1}",
        color=colors[i + 2],   # offset by 2 to avoid reusing train/test colours
        linewidth=1.4,
        alpha=0.85,
    )

plt.title("5 independent fits  -  MLP([1, 2, 1])", fontsize=12)
plt.xlabel("x")
plt.ylabel("y")
plt.legend(
    loc="upper left",
    bbox_to_anchor=(1.01, 1),   # place legend outside the axes on the right
    borderaxespad=0,
    fontsize=9,
)
plt.tight_layout()
saveplot("a01_2_repeated-fits.pdf")
```

```
--- Run 1 ---
X1 shape: torch.Size([100, 1])
Repetition  0:

/home/tu2/anaconda3/envs/deep-learning/lib/python3.13/site-
packages/scipy/optimize/_minimize.py:779: OptimizeWarning: Desired error not
necessarily achieved due to precision loss.
  res = _minimize_bfgs(fun, x0, args, jac, callback, **options)

         Current function value: 0.286909
         Iterations: 318
         Function evaluations: 449
         Gradient evaluations: 439
```

```
best_cost=0.287

--- Run 2 ---
X1 shape: torch.Size([100, 1])
Repetition  0:

/home/tu2/anaconda3/envs/deep-learning/lib/python3.13/site-
packages/scipy/optimize/_minimize.py:779: OptimizeWarning: Desired error not
necessarily achieved due to precision loss.
  res = _minimize_bfgs(fun, x0, args, jac, callback, **options)

         Current function value: 0.079572
         Iterations: 406
         Function evaluations: 568
         Gradient evaluations: 557
best_cost=0.080

--- Run 3 ---
X1 shape: torch.Size([100, 1])
Repetition  0: Optimization terminated successfully.
         Current function value: 0.302737
         Iterations: 138
         Function evaluations: 240
         Gradient evaluations: 229
best_cost=0.303

--- Run 4 ---
X1 shape: torch.Size([100, 1])
Repetition  0:

/home/tu2/anaconda3/envs/deep-learning/lib/python3.13/site-
packages/scipy/optimize/_minimize.py:779: OptimizeWarning: Desired error not
necessarily achieved due to precision loss.
  res = _minimize_bfgs(fun, x0, args, jac, callback, **options)

         Current function value: 0.079572
         Iterations: 407
         Function evaluations: 562
         Gradient evaluations: 555
best_cost=0.080

--- Run 5 ---
X1 shape: torch.Size([100, 1])
Repetition  0:          Current function value: 0.079572
         Iterations: 446
         Function evaluations: 701
         Gradient evaluations: 688
best_cost=0.080

/home/tu2/anaconda3/envs/deep-learning/lib/python3.13/site-
```

```
packages/scipy/optimize/_minimize.py:779: OptimizeWarning: Desired error not
necessarily achieved due to precision loss.
  res = _minimize_bfgs(fun, x0, args, jac, callback, **options)
```



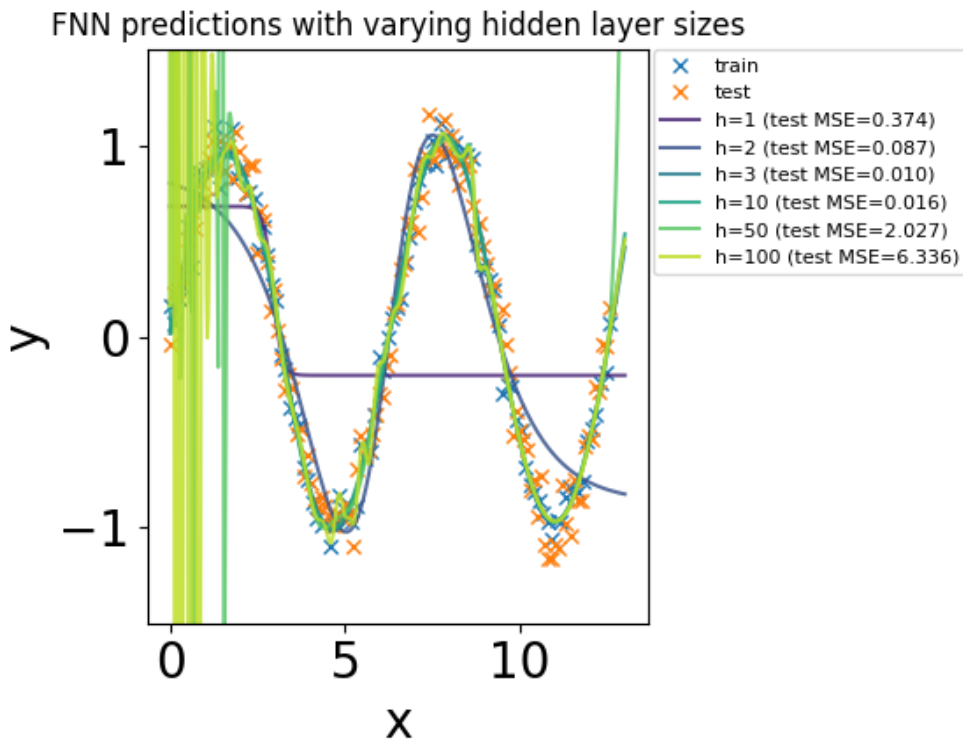
1.3 2c Width

[21]:
```
# Experiment with different hidden layer sizes. To avoid recomputing
# models, you may want to save your models using torch.save(model, filename) and
# load them again using torch.load(filename).
# TODO: YOUR CODE HERE
hidden_sizes_list = [1, 2, 3, 10, 50, 100]
models = {}
train_mses = {}
test_mses = {}

MODEL_DIR = os.path.join(os.getcwd(), "models")
os.makedirs(MODEL_DIR, exist_ok=True)

for h in hidden_sizes_list:
    fname = os.path.join(MODEL_DIR, f"a01_2c_model_h{h}.pt")
    if os.path.exists(fname):
```

```
        print(f"=== Loading cached model  h={h} from {fname} ===")
        model_h = torch.load(fname, weights_only=False)
    else:
        print(f"\n=== Training MLP  h={h} ===")
        model_h = train1([h], nreps=10)
        torch.save(model_h, fname)
        print(f"    Saved to {fname}")
    models[h] = model_h

    train_mse = F.mse_loss(y1, model_h(X1)).item()
    test_mse  = F.mse_loss(y1test, model_h(X1test)).item()
    train_mses[h] = train_mse
    test_mses[h]  = test_mse
    print(f"  Train MSE: {train_mse:.6f}  |  Test MSE: {test_mse:.6f}")

# Print summary table
print("\n" + "=" * 50)
print(f"{'Hidden Units':<15} {'Train MSE':<15} {'Test MSE':<15}")
print("=" * 50)
for h in hidden_sizes_list:
    print(f"{h:<15} {train_mses[h]:<15.6f} {test_mses[h]:<15.6f}")
print("=" * 50)

# Plot all predictions in a single plot
nextplot()
plot1(X1, y1, label="train")
plot1(X1test, y1test, label="test")

X_plot = torch.linspace(0, 13, 500).unsqueeze(1)
colors = plt.cm.viridis(np.linspace(0.1, 0.9, len(hidden_sizes_list)))

for i, h in enumerate(hidden_sizes_list):
    y_pred = models[h](X_plot).detach().numpy()
    plt.plot(
        X_plot.numpy(),
        y_pred,
        label=f"h={h} (test MSE={test_mses[h]:.3f})",
        color=colors[i],
        linewidth=1.5,
        alpha=0.85,
    )

plt.title("FNN predictions with varying hidden layer sizes", fontsize=12)
plt.xlabel("x")
plt.ylabel("y")
plt.ylim(-1.5, 1.5)   # clamp y-axis to the data range; large-h spikes would␣
 ↪otherwise squash the view
```

```python
plt.legend(
    loc="upper left",
    bbox_to_anchor=(1.01, 1),
    borderaxespad=0,
    fontsize=8,
)
plt.tight_layout()
saveplot("a01_2_width-comparison.pdf")
```

```
=== Loading cached model  h=1 from /home/tu2/projects/IE678-Deep-
Learning/a01-fnn/python/models/a01_2c_model_h1.pt ===
  Train MSE: 0.372919  |  Test MSE: 0.374317
=== Loading cached model  h=2 from /home/tu2/projects/IE678-Deep-
Learning/a01-fnn/python/models/a01_2c_model_h2.pt ===
  Train MSE: 0.079571  |  Test MSE: 0.086714
=== Loading cached model  h=3 from /home/tu2/projects/IE678-Deep-
Learning/a01-fnn/python/models/a01_2c_model_h3.pt ===
  Train MSE: 0.007324  |  Test MSE: 0.010335
=== Loading cached model  h=10 from /home/tu2/projects/IE678-Deep-
Learning/a01-fnn/python/models/a01_2c_model_h10.pt ===
  Train MSE: 0.005916  |  Test MSE: 0.015976
=== Loading cached model  h=50 from /home/tu2/projects/IE678-Deep-
Learning/a01-fnn/python/models/a01_2c_model_h50.pt ===
  Train MSE: 0.003007  |  Test MSE: 2.026847
=== Loading cached model  h=100 from /home/tu2/projects/IE678-Deep-
Learning/a01-fnn/python/models/a01_2c_model_h100.pt ===
  Train MSE: 0.002126  |  Test MSE: 6.336472

==================================================
Hidden Units    Train MSE       Test MSE
==================================================
1               0.372919        0.374317
2               0.079571        0.086714
3               0.007324        0.010335
10              0.005916        0.015976
50              0.003007        2.026847
100             0.002126        6.336472
==================================================
```



1.4 2d Distributed representations

[22]:
```
# train a model to analyze
model = train1([2])

# TODO: YOUR CODE HERE
```

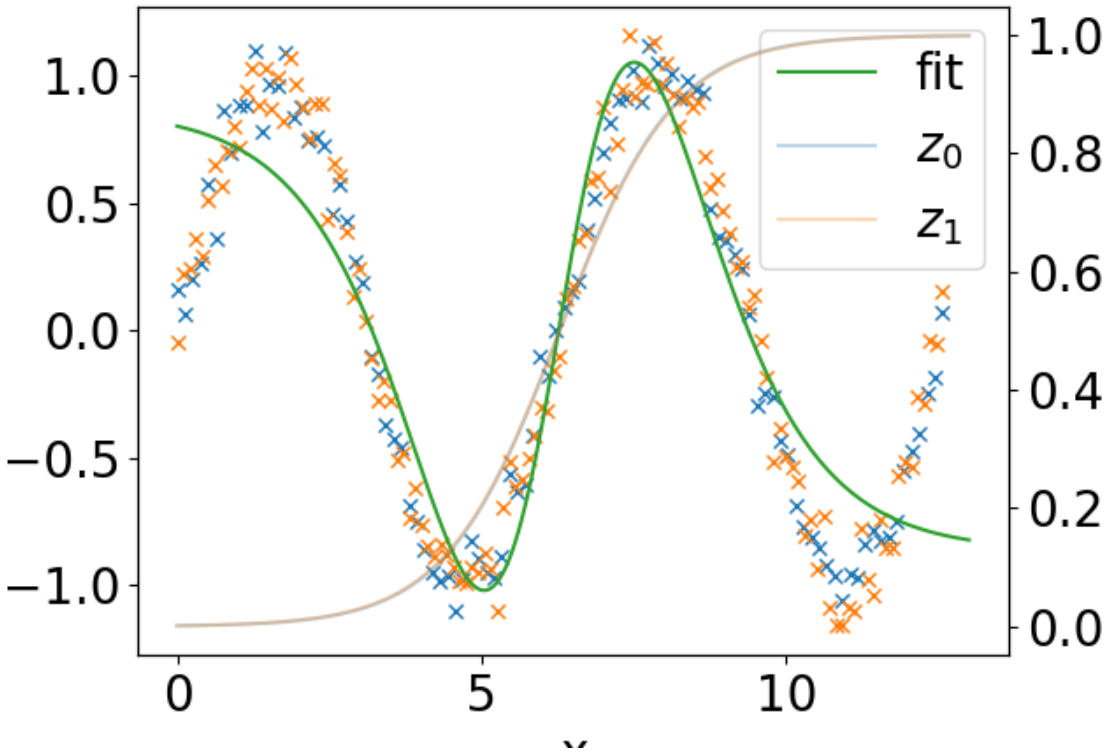
```
X1 shape: torch.Size([100, 1])
Repetition  0:          Current function value: 0.286918
         Iterations: 140
         Function evaluations: 271
         Gradient evaluations: 258
best_cost=0.287
X1 shape: torch.Size([100, 1])
Repetition  1:
```

```
/home/tu2/anaconda3/envs/deep-learning/lib/python3.13/site-
packages/scipy/optimize/_minimize.py:779: OptimizeWarning: Desired error not
necessarily achieved due to precision loss.
  res = _minimize_bfgs(fun, x0, args, jac, callback, **options)
/home/tu2/anaconda3/envs/deep-learning/lib/python3.13/site-
```

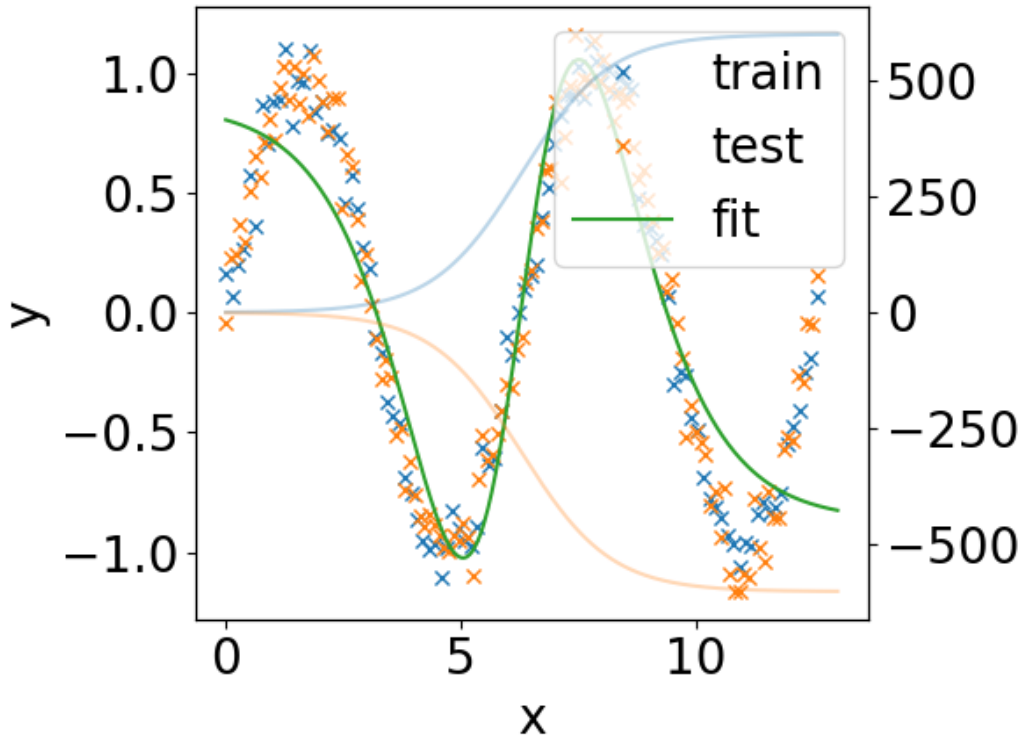
```
packages/scipy/optimize/_minimize.py:779: OptimizeWarning: Desired error not
necessarily achieved due to precision loss.
  res = _minimize_bfgs(fun, x0, args, jac, callback, **options)

         Current function value: 0.079573
         Iterations: 354
         Function evaluations: 471
         Gradient evaluations: 461
best_cost=0.080
X1 shape: torch.Size([100, 1])
Repetition  2:

/home/tu2/anaconda3/envs/deep-learning/lib/python3.13/site-
packages/scipy/optimize/_minimize.py:779: OptimizeWarning: Desired error not
necessarily achieved due to precision loss.
  res = _minimize_bfgs(fun, x0, args, jac, callback, **options)
/home/tu2/anaconda3/envs/deep-learning/lib/python3.13/site-
packages/scipy/optimize/_minimize.py:779: OptimizeWarning: Desired error not
necessarily achieved due to precision loss.
  res = _minimize_bfgs(fun, x0, args, jac, callback, **options)

         Current function value: 0.286909
         Iterations: 475
         Function evaluations: 674
         Gradient evaluations: 662
best_cost=0.080
X1 shape: torch.Size([100, 1])
Repetition  3:          Current function value: 0.079573
         Iterations: 359
         Function evaluations: 480
         Gradient evaluations: 471
best_cost=0.080
X1 shape: torch.Size([100, 1])
Repetition  4: Optimization terminated successfully.
         Current function value: 0.277769
         Iterations: 104
         Function evaluations: 139
         Gradient evaluations: 139
best_cost=0.080
X1 shape: torch.Size([100, 1])
Repetition  5:          Current function value: 0.079573
         Iterations: 359
         Function evaluations: 492
         Gradient evaluations: 483
best_cost=0.080
X1 shape: torch.Size([100, 1])
Repetition  6:

/home/tu2/anaconda3/envs/deep-learning/lib/python3.13/site-
packages/scipy/optimize/_minimize.py:779: OptimizeWarning: Desired error not
```

```
necessarily achieved due to precision loss.
  res = _minimize_bfgs(fun, x0, args, jac, callback, **options)

Optimization terminated successfully.
         Current function value: 0.372919
         Iterations: 83
         Function evaluations: 87
         Gradient evaluations: 87
best_cost=0.080
X1 shape: torch.Size([100, 1])
Repetition  7: Optimization terminated successfully.
         Current function value: 0.357250
         Iterations: 81
         Function evaluations: 87
         Gradient evaluations: 87
best_cost=0.080
X1 shape: torch.Size([100, 1])
Repetition  8: Optimization terminated successfully.
         Current function value: 0.372709
         Iterations: 78
         Function evaluations: 85
         Gradient evaluations: 85
best_cost=0.080
X1 shape: torch.Size([100, 1])
Repetition  9:          Current function value: 0.079572
         Iterations: 403
         Function evaluations: 552
         Gradient evaluations: 542
best_cost=0.080

/home/tu2/anaconda3/envs/deep-learning/lib/python3.13/site-
packages/scipy/optimize/_minimize.py:779: OptimizeWarning: Desired error not
necessarily achieved due to precision loss.
  res = _minimize_bfgs(fun, x0, args, jac, callback, **options)
```

[23]:
```python
# plot the fit as well as the outputs of each neuron in the hidden
# layer (scale for the latter is shown on right y-axis)
nextplot()
plot1(X1, y1, label="train")
plot1(X1test, y1test, label="test")
plot1fit(torch.linspace(0, 13, 500).unsqueeze(1), model, hidden=True,␣
 ↪scale=False)
saveplot("a01_2_distributed-reps.pdf")
```



[24]:
```
# plot the fit as well as the outputs of each neuron in the hidden layer, scaled
# by its weight for the output neuron (scale for the latter is shown on right
# y-axis)
nextplot()
plot1(X1, y1, label="train")
plot1(X1test, y1test, label="test")
plot1fit(torch.linspace(0, 13, 500).unsqueeze(1), model, hidden=True,␣
↪scale=True)
plt.legend(loc="upper right")
plt.tight_layout()
saveplot("a01_2_distributed-reps-scaled.pdf")
```



[25]:
```
# 2d(iii): Repeat (i) and (ii) with 3 hidden neurons
# TODO: YOUR CODE HERE
X_dense = torch.linspace(0, 13, 500).unsqueeze(1)

model_h3 = train1([3])

nextplot()
plot1(X1, y1, label="train")
plot1(X1test, y1test, label="test")
plot1fit(X_dense, model_h3, hidden=True, scale=False)
saveplot("a01_2_distributed-reps-h3.pdf")

nextplot()
plot1(X1, y1, label="train")
plot1(X1test, y1test, label="test")
plot1fit(X_dense, model_h3, hidden=True, scale=True)
plt.legend(loc="upper right")
plt.tight_layout()
saveplot("a01_2_distributed-reps-h3-scaled.pdf")
```

```
X1 shape: torch.Size([100, 1])
```

```
Repetition  0:

/home/tu2/anaconda3/envs/deep-learning/lib/python3.13/site-
packages/scipy/optimize/_minimize.py:779: OptimizeWarning: Desired error not
necessarily achieved due to precision loss.
  res = _minimize_bfgs(fun, x0, args, jac, callback, **options)

         Current function value: 0.051869
         Iterations: 337
         Function evaluations: 498
         Gradient evaluations: 487
best_cost=0.052
X1 shape: torch.Size([100, 1])
Repetition  1:

/home/tu2/anaconda3/envs/deep-learning/lib/python3.13/site-
packages/scipy/optimize/_minimize.py:779: OptimizeWarning: Desired error not
necessarily achieved due to precision loss.
  res = _minimize_bfgs(fun, x0, args, jac, callback, **options)

         Current function value: 0.007324
         Iterations: 564
         Function evaluations: 806
         Gradient evaluations: 792
best_cost=0.007
X1 shape: torch.Size([100, 1])
Repetition  2:

/home/tu2/anaconda3/envs/deep-learning/lib/python3.13/site-
packages/scipy/optimize/_minimize.py:779: OptimizeWarning: Desired error not
necessarily achieved due to precision loss.
  res = _minimize_bfgs(fun, x0, args, jac, callback, **options)

         Current function value: 0.049878
         Iterations: 473
         Function evaluations: 620
         Gradient evaluations: 609
best_cost=0.007
X1 shape: torch.Size([100, 1])
Repetition  3:

/home/tu2/anaconda3/envs/deep-learning/lib/python3.13/site-
packages/scipy/optimize/_minimize.py:779: OptimizeWarning: Desired error not
necessarily achieved due to precision loss.
  res = _minimize_bfgs(fun, x0, args, jac, callback, **options)
/home/tu2/anaconda3/envs/deep-learning/lib/python3.13/site-
packages/scipy/optimize/_minimize.py:779: OptimizeWarning: Desired error not
necessarily achieved due to precision loss.
  res = _minimize_bfgs(fun, x0, args, jac, callback, **options)

         Current function value: 0.075173
         Iterations: 424
```

```
         Function evaluations: 588
         Gradient evaluations: 579
best_cost=0.007
X1 shape: torch.Size([100, 1])
Repetition  4:          Current function value: 0.275501
         Iterations: 240
         Function evaluations: 332
         Gradient evaluations: 321
best_cost=0.007
X1 shape: torch.Size([100, 1])
Repetition  5:
```

```
/home/tu2/anaconda3/envs/deep-learning/lib/python3.13/site-
packages/scipy/optimize/_minimize.py:779: OptimizeWarning: Desired error not
necessarily achieved due to precision loss.
  res = _minimize_bfgs(fun, x0, args, jac, callback, **options)
/home/tu2/anaconda3/envs/deep-learning/lib/python3.13/site-
packages/scipy/optimize/_minimize.py:779: OptimizeWarning: Desired error not
necessarily achieved due to precision loss.
  res = _minimize_bfgs(fun, x0, args, jac, callback, **options)
```

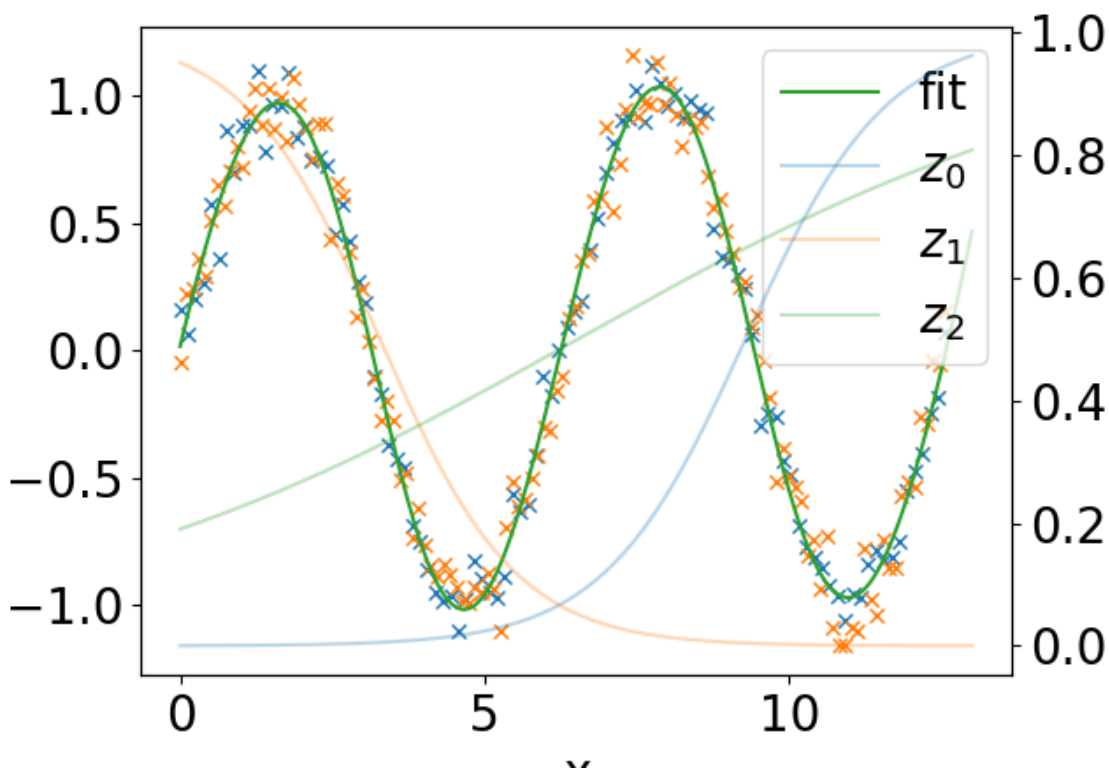
```
         Current function value: 0.007324
         Iterations: 602
         Function evaluations: 737
         Gradient evaluations: 726
best_cost=0.007
X1 shape: torch.Size([100, 1])
Repetition  6:          Current function value: 0.060586
         Iterations: 296
         Function evaluations: 462
         Gradient evaluations: 447
best_cost=0.007
X1 shape: torch.Size([100, 1])
Repetition  7:
```

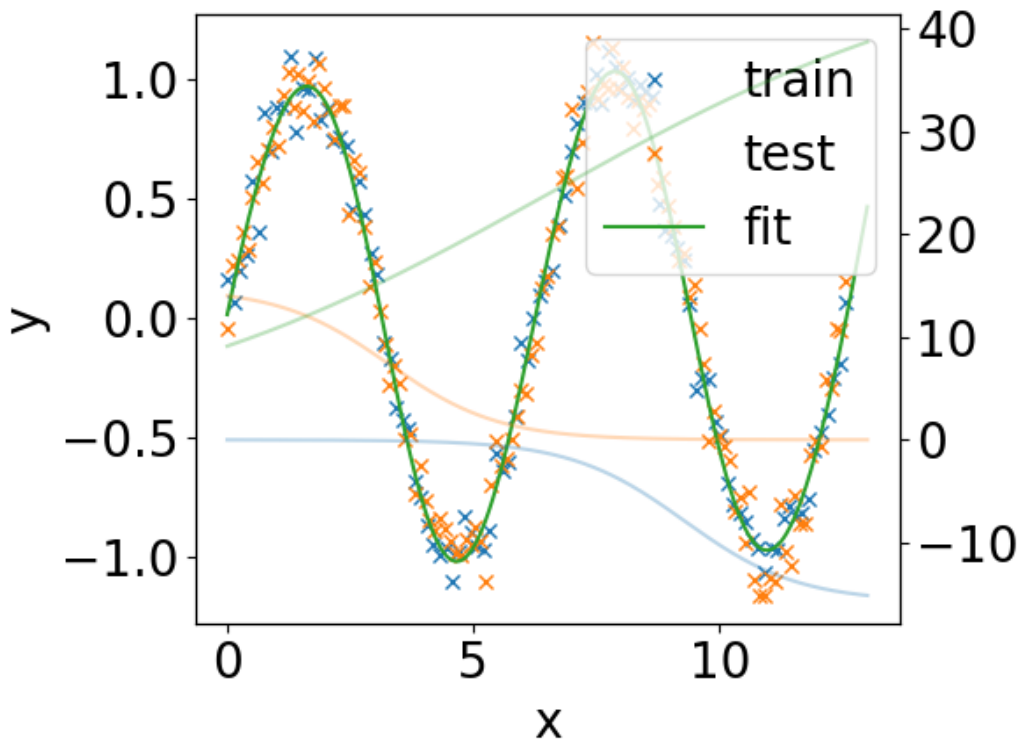
```
/home/tu2/anaconda3/envs/deep-learning/lib/python3.13/site-
packages/scipy/optimize/_minimize.py:779: OptimizeWarning: Desired error not
necessarily achieved due to precision loss.
  res = _minimize_bfgs(fun, x0, args, jac, callback, **options)
```

```
         Current function value: 0.051891
         Iterations: 439
         Function evaluations: 665
         Gradient evaluations: 653
best_cost=0.007
X1 shape: torch.Size([100, 1])
Repetition  8:
```

```
/home/tu2/anaconda3/envs/deep-learning/lib/python3.13/site-
packages/scipy/optimize/_minimize.py:779: OptimizeWarning: Desired error not
```

```
necessarily achieved due to precision loss.
  res = _minimize_bfgs(fun, x0, args, jac, callback, **options)

         Current function value: 0.049877
         Iterations: 525
         Function evaluations: 744
         Gradient evaluations: 733
best_cost=0.007
X1 shape: torch.Size([100, 1])
Repetition  9:

/home/tu2/anaconda3/envs/deep-learning/lib/python3.13/site-
packages/scipy/optimize/_minimize.py:779: OptimizeWarning: Desired error not
necessarily achieved due to precision loss.
  res = _minimize_bfgs(fun, x0, args, jac, callback, **options)

         Current function value: 0.049888
         Iterations: 381
         Function evaluations: 515
         Gradient evaluations: 503
best_cost=0.007
```





[26]:
```
# 2d(iii): Repeat (i) and (ii) with 10 hidden neurons
model_h10 = train1([10])

nextplot()
plot1(X1, y1, label="train")
plot1(X1test, y1test, label="test")
plot1fit(X_dense, model_h10, hidden=True, scale=False)
saveplot("a01_2_distributed-reps-h10.pdf")

nextplot()
plot1(X1, y1, label="train")
plot1(X1test, y1test, label="test")
plot1fit(X_dense, model_h10, hidden=True, scale=True)
plt.legend(loc="upper right")
plt.tight_layout()
saveplot("a01_2_distributed-reps-h10-scaled.pdf")
```

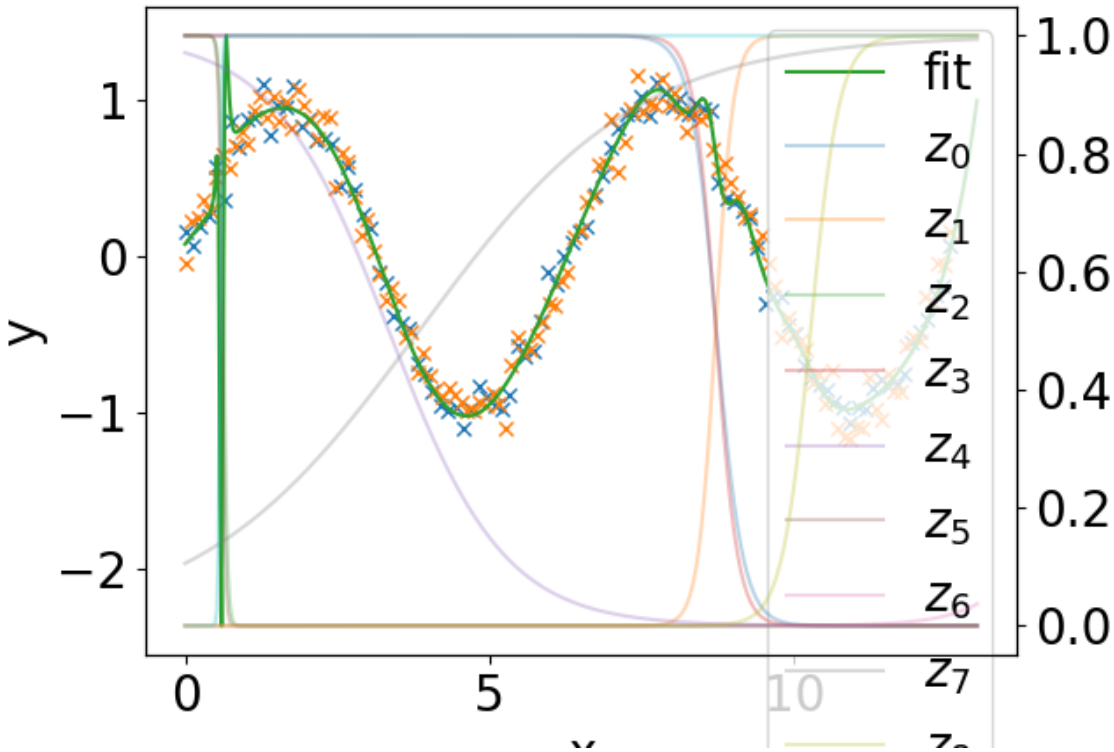
```
X1 shape: torch.Size([100, 1])
Repetition  0:
```

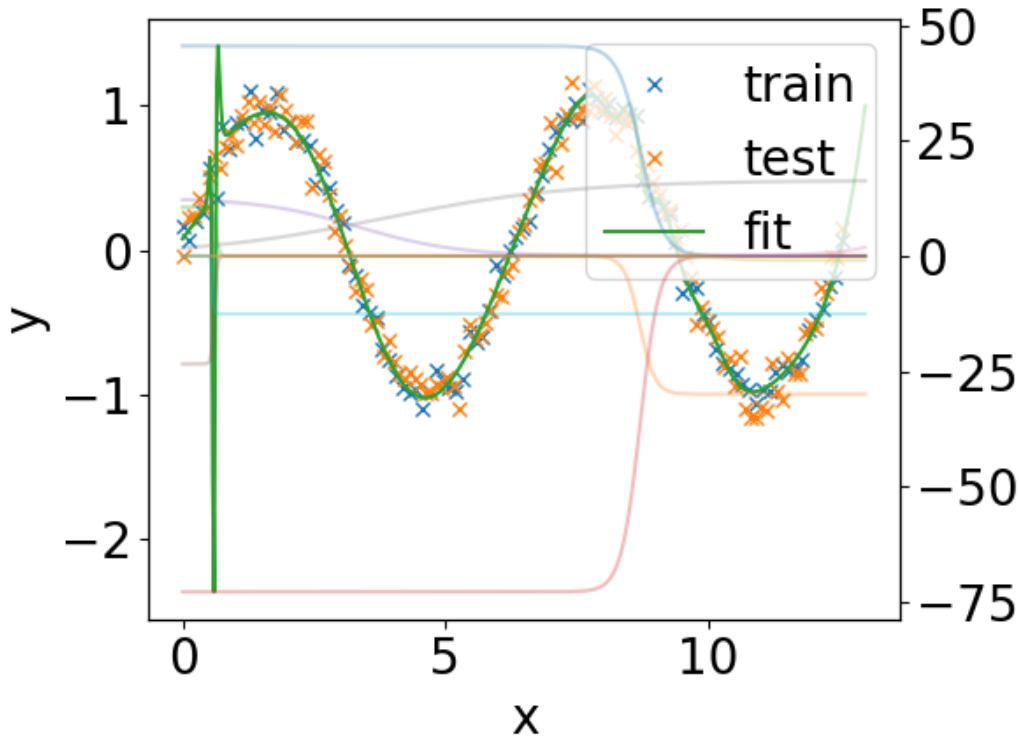
```
/home/tu2/anaconda3/envs/deep-learning/lib/python3.13/site-
```

```
packages/scipy/optimize/_minimize.py:779: OptimizeWarning: Desired error not
necessarily achieved due to precision loss.
  res = _minimize_bfgs(fun, x0, args, jac, callback, **options)

         Current function value: 0.004748
         Iterations: 1600
         Function evaluations: 1785
         Gradient evaluations: 1775
best_cost=0.005
X1 shape: torch.Size([100, 1])
Repetition  1:

/home/tu2/anaconda3/envs/deep-learning/lib/python3.13/site-
packages/scipy/optimize/_minimize.py:779: OptimizeWarning: Desired error not
necessarily achieved due to precision loss.
  res = _minimize_bfgs(fun, x0, args, jac, callback, **options)

         Current function value: 0.006553
         Iterations: 1357
         Function evaluations: 1577
         Gradient evaluations: 1566
best_cost=0.005
X1 shape: torch.Size([100, 1])
Repetition  2:

/home/tu2/anaconda3/envs/deep-learning/lib/python3.13/site-
packages/scipy/optimize/_minimize.py:779: OptimizeWarning: Desired error not
necessarily achieved due to precision loss.
  res = _minimize_bfgs(fun, x0, args, jac, callback, **options)

         Current function value: 0.006653
         Iterations: 1171
         Function evaluations: 1356
         Gradient evaluations: 1346
best_cost=0.005
X1 shape: torch.Size([100, 1])
Repetition  3:

/home/tu2/anaconda3/envs/deep-learning/lib/python3.13/site-
packages/scipy/optimize/_minimize.py:779: OptimizeWarning: Desired error not
necessarily achieved due to precision loss.
  res = _minimize_bfgs(fun, x0, args, jac, callback, **options)

         Current function value: 0.006659
         Iterations: 1523
         Function evaluations: 1826
         Gradient evaluations: 1813
best_cost=0.005
X1 shape: torch.Size([100, 1])
Repetition  4:
```

```
/home/tu2/anaconda3/envs/deep-learning/lib/python3.13/site-
packages/scipy/optimize/_minimize.py:779: OptimizeWarning: Desired error not
necessarily achieved due to precision loss.
  res = _minimize_bfgs(fun, x0, args, jac, callback, **options)

         Current function value: 0.005853
         Iterations: 1641
         Function evaluations: 1852
         Gradient evaluations: 1840
best_cost=0.005
X1 shape: torch.Size([100, 1])
Repetition  5:

/home/tu2/anaconda3/envs/deep-learning/lib/python3.13/site-
packages/scipy/optimize/_minimize.py:779: OptimizeWarning: Desired error not
necessarily achieved due to precision loss.
  res = _minimize_bfgs(fun, x0, args, jac, callback, **options)

         Current function value: 0.006561
         Iterations: 1674
         Function evaluations: 1954
         Gradient evaluations: 1944
best_cost=0.005
X1 shape: torch.Size([100, 1])
Repetition  6:

/home/tu2/anaconda3/envs/deep-learning/lib/python3.13/site-
packages/scipy/optimize/_minimize.py:779: OptimizeWarning: Desired error not
necessarily achieved due to precision loss.
  res = _minimize_bfgs(fun, x0, args, jac, callback, **options)

         Current function value: 0.005799
         Iterations: 2652
         Function evaluations: 3131
         Gradient evaluations: 3113
best_cost=0.005
X1 shape: torch.Size([100, 1])
Repetition  7:

/home/tu2/anaconda3/envs/deep-learning/lib/python3.13/site-
packages/scipy/optimize/_minimize.py:779: OptimizeWarning: Desired error not
necessarily achieved due to precision loss.
  res = _minimize_bfgs(fun, x0, args, jac, callback, **options)

         Current function value: 0.006599
         Iterations: 2637
         Function evaluations: 3053
         Gradient evaluations: 3041
best_cost=0.005
X1 shape: torch.Size([100, 1])
Repetition  8:
```

```
/home/tu2/anaconda3/envs/deep-learning/lib/python3.13/site-
packages/scipy/optimize/_minimize.py:779: OptimizeWarning: Desired error not
necessarily achieved due to precision loss.
  res = _minimize_bfgs(fun, x0, args, jac, callback, **options)

         Current function value: 0.006658
         Iterations: 1095
         Function evaluations: 1280
         Gradient evaluations: 1268
best_cost=0.005
X1 shape: torch.Size([100, 1])
Repetition  9:

/home/tu2/anaconda3/envs/deep-learning/lib/python3.13/site-
packages/scipy/optimize/_minimize.py:779: OptimizeWarning: Desired error not
necessarily achieved due to precision loss.
  res = _minimize_bfgs(fun, x0, args, jac, callback, **options)
/home/tu2/projects/IE678-Deep-Learning/a01-fnn/python/a01_helper.py:40:
RuntimeWarning: More than 20 figures have been opened. Figures created through
the pyplot interface (`matplotlib.pyplot.figure`) are retained until explicitly
closed and may consume too much memory. (To control this warning, see the
rcParam `figure.max_open_warning`). Consider using `matplotlib.pyplot.close()`.
  plt.figure()  # this creates a new plot

         Current function value: 0.005940
         Iterations: 2646
         Function evaluations: 2998
         Gradient evaluations: 2986
best_cost=0.005
```


fit
z_0
z_1
z_2
z_3
z_4
z_5
z_6
z_7
y
1
0
−1
−2
0
5
10
1.0
0.8
0.6
0.4
0.2
0.0



1.5 2e Experiment with different optimizers (optional)

[27]:
```
# PyTorch provides many gradient-based optimizers; see
# https://pytorch.org/docs/stable/optim.html. You can use a PyTorch optimizer
# as follows.
train_adam = lambda model, **kwargs: fnn_train(
    X1, y1, model, optimizer=torch.optim.Adam(model.parameters(), lr=0.01),␣
 ↪**kwargs
)
model = train1([50], nreps=1, train=train_adam, max_epochs=5000, tol=1e-8,␣
 ↪verbose=True)
```

```
X1 shape: torch.Size([100, 1])
Repetition  0: Epoch     0: cost=   3.401
Epoch     1: cost=   2.482
Epoch     2: cost=   1.739
Epoch     3: cost=   1.174
Epoch     4: cost=   0.782
Epoch     5: cost=   0.552
Epoch     6: cost=   0.461
Epoch     7: cost=   0.480
```

```
Epoch     8: cost=   0.567
Epoch     9: cost=   0.680
Epoch    10: cost=   0.781
Epoch    11: cost=   0.845
Epoch    12: cost=   0.863
Epoch    13: cost=   0.837
Epoch    14: cost=   0.780
Epoch    15: cost=   0.703
Epoch    16: cost=   0.623
Epoch    17: cost=   0.550
Epoch    18: cost=   0.492
Epoch    19: cost=   0.455
Epoch    20: cost=   0.438
Epoch    21: cost=   0.438
Epoch    22: cost=   0.452
Epoch    23: cost=   0.473
Epoch    24: cost=   0.494
Epoch    25: cost=   0.511
Epoch    26: cost=   0.521
Epoch    27: cost=   0.521
Epoch    28: cost=   0.514
Epoch    29: cost=   0.500
Epoch    30: cost=   0.483
Epoch    31: cost=   0.465
Epoch    32: cost=   0.449
Epoch    33: cost=   0.438
Epoch    34: cost=   0.432
Epoch    35: cost=   0.431
Epoch    36: cost=   0.433
Epoch    37: cost=   0.438
Epoch    38: cost=   0.444
Epoch    39: cost=   0.449
Epoch    40: cost=   0.451
Epoch    41: cost=   0.451
Epoch    42: cost=   0.449
Epoch    43: cost=   0.445
Epoch    44: cost=   0.440
Epoch    45: cost=   0.435
Epoch    46: cost=   0.431
Epoch    47: cost=   0.429
Epoch    48: cost=   0.427
Epoch    49: cost=   0.428
Epoch    50: cost=   0.429
Epoch    51: cost=   0.430
Epoch    52: cost=   0.431
Epoch    53: cost=   0.432
Epoch    54: cost=   0.432
Epoch    55: cost=   0.431
```

```
Epoch    56: cost=   0.430
Epoch    57: cost=   0.428
Epoch    58: cost=   0.427
Epoch    59: cost=   0.425
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Epoch    61: cost=   0.424
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Epoch   100: cost=   0.416
Epoch   101: cost=   0.416
Epoch   102: cost=   0.416
Epoch   103: cost=   0.415
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Epoch   104: cost=   0.415
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Epoch   200: cost=   0.390
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Epoch  4513: cost=   0.007
Epoch  4514: cost=   0.007
Epoch  4515: cost=   0.007
Epoch  4516: cost=   0.007
Epoch  4517: cost=   0.007
Epoch  4518: cost=   0.007
Epoch  4519: cost=   0.007
```

```
Epoch  4520: cost=   0.007
Epoch  4521: cost=   0.007
Epoch  4522: cost=   0.007
Epoch  4523: cost=   0.007
Epoch  4524: cost=   0.007
Epoch  4525: cost=   0.007
Epoch  4526: cost=   0.007
Epoch  4527: cost=   0.007
Epoch  4528: cost=   0.007
Epoch  4529: cost=   0.007
Epoch  4530: cost=   0.007
Epoch  4531: cost=   0.007
Epoch  4532: cost=   0.007
Epoch  4533: cost=   0.007
Epoch  4534: cost=   0.007
Epoch  4535: cost=   0.007
Epoch  4536: cost=   0.007
Epoch  4537: cost=   0.007
Epoch  4538: cost=   0.007
Epoch  4539: cost=   0.007
Epoch  4540: cost=   0.007
Epoch  4541: cost=   0.007
Epoch  4542: cost=   0.007
Epoch  4543: cost=   0.007
Epoch  4544: cost=   0.007
Epoch  4545: cost=   0.007
Epoch  4546: cost=   0.007
Epoch  4547: cost=   0.007
Epoch  4548: cost=   0.007
Epoch  4549: cost=   0.007
Epoch  4550: cost=   0.007
Epoch  4551: cost=   0.007
Epoch  4552: cost=   0.007
Epoch  4553: cost=   0.007
Epoch  4554: cost=   0.007
Epoch  4555: cost=   0.007
Epoch  4556: cost=   0.007
Epoch  4557: cost=   0.007
Epoch  4558: cost=   0.007
Epoch  4559: cost=   0.007
Epoch  4560: cost=   0.007
Epoch  4561: cost=   0.007
Epoch  4562: cost=   0.007
Epoch  4563: cost=   0.007
Epoch  4564: cost=   0.007
Epoch  4565: cost=   0.007
Epoch  4566: cost=   0.007
Epoch  4567: cost=   0.007
```

```
Epoch  4568: cost=   0.007
Epoch  4569: cost=   0.007
best_cost=0.007
```

[28]:
```
# Experiment with different number of layers and activation functions. Here is
# an example with three hidden layers (of sizes 4, 5, and 6) and ReLU␣
 ↪activations.
#
# You can also plot the outputs of the hidden neurons in the first layer (using
# the same code above).
model = train1([4, 5, 6], nreps=50, phi=F.relu)
nextplot()
plot1(X1, y1, label="train")
plot1(X1test, y1test, label="test")
plot1fit(torch.linspace(0, 13, 500).unsqueeze(1), model)
print("Training error:", F.mse_loss(y1, model(X1)).item())
print("Test error    :", F.mse_loss(y1test, model(X1test)).item())
```

```
X1 shape: torch.Size([100, 1])
Repetition  0: Optimization terminated successfully.
         Current function value: 0.506238
         Iterations: 3
         Function evaluations: 5
         Gradient evaluations: 5
best_cost=0.506
X1 shape: torch.Size([100, 1])
Repetition  1:

         Current function value: 0.356164
         Iterations: 57
         Function evaluations: 139
         Gradient evaluations: 135
best_cost=0.356
X1 shape: torch.Size([100, 1])
Repetition  2:          Current function value: 0.506139
         Iterations: 31
         Function evaluations: 123
         Gradient evaluations: 120
best_cost=0.356
X1 shape: torch.Size([100, 1])
Repetition  3:

/home/tu2/anaconda3/envs/deep-learning/lib/python3.13/site-
packages/scipy/optimize/_minimize.py:779: OptimizeWarning: Desired error not
necessarily achieved due to precision loss.
  res = _minimize_bfgs(fun, x0, args, jac, callback, **options)
/home/tu2/anaconda3/envs/deep-learning/lib/python3.13/site-
packages/scipy/optimize/_minimize.py:779: OptimizeWarning: Desired error not
necessarily achieved due to precision loss.
```

```
  res = _minimize_bfgs(fun, x0, args, jac, callback, **options)
/home/tu2/anaconda3/envs/deep-learning/lib/python3.13/site-
packages/scipy/optimize/_minimize.py:779: OptimizeWarning: Desired error not
necessarily achieved due to precision loss.
  res = _minimize_bfgs(fun, x0, args, jac, callback, **options)

         Current function value: 0.012184
         Iterations: 83
         Function evaluations: 147
         Gradient evaluations: 144
best_cost=0.012
X1 shape: torch.Size([100, 1])
Repetition  4:          Current function value: 0.355556
         Iterations: 59
         Function evaluations: 141
         Gradient evaluations: 137
best_cost=0.012
X1 shape: torch.Size([100, 1])
Repetition  5:          Current function value: 0.271873
         Iterations: 38
         Function evaluations: 116
         Gradient evaluations: 110
best_cost=0.012
X1 shape: torch.Size([100, 1])
Repetition  6:

/home/tu2/anaconda3/envs/deep-learning/lib/python3.13/site-
packages/scipy/optimize/_minimize.py:779: OptimizeWarning: Desired error not
necessarily achieved due to precision loss.
  res = _minimize_bfgs(fun, x0, args, jac, callback, **options)
/home/tu2/anaconda3/envs/deep-learning/lib/python3.13/site-
packages/scipy/optimize/_minimize.py:779: OptimizeWarning: Desired error not
necessarily achieved due to precision loss.
  res = _minimize_bfgs(fun, x0, args, jac, callback, **options)
/home/tu2/anaconda3/envs/deep-learning/lib/python3.13/site-
packages/scipy/optimize/_minimize.py:779: OptimizeWarning: Desired error not
necessarily achieved due to precision loss.
  res = _minimize_bfgs(fun, x0, args, jac, callback, **options)
/home/tu2/anaconda3/envs/deep-learning/lib/python3.13/site-
packages/scipy/optimize/_minimize.py:779: OptimizeWarning: Desired error not
necessarily achieved due to precision loss.
  res = _minimize_bfgs(fun, x0, args, jac, callback, **options)

         Current function value: 0.011510
         Iterations: 185
         Function evaluations: 308
         Gradient evaluations: 296
best_cost=0.012
X1 shape: torch.Size([100, 1])
```

```
Repetition  7: Optimization terminated successfully.
         Current function value: 0.506238
         Iterations: 9
         Function evaluations: 14
         Gradient evaluations: 14
best_cost=0.012
X1 shape: torch.Size([100, 1])
Repetition  8:          Current function value: 0.355945
         Iterations: 103
         Function evaluations: 201
         Gradient evaluations: 197
best_cost=0.012
X1 shape: torch.Size([100, 1])
Repetition  9:

/home/tu2/anaconda3/envs/deep-learning/lib/python3.13/site-
packages/scipy/optimize/_minimize.py:779: OptimizeWarning: Desired error not
necessarily achieved due to precision loss.
  res = _minimize_bfgs(fun, x0, args, jac, callback, **options)
/home/tu2/anaconda3/envs/deep-learning/lib/python3.13/site-
packages/scipy/optimize/_minimize.py:779: OptimizeWarning: Desired error not
necessarily achieved due to precision loss.
  res = _minimize_bfgs(fun, x0, args, jac, callback, **options)
/home/tu2/anaconda3/envs/deep-learning/lib/python3.13/site-
packages/scipy/optimize/_minimize.py:779: OptimizeWarning: Desired error not
necessarily achieved due to precision loss.
  res = _minimize_bfgs(fun, x0, args, jac, callback, **options)

         Current function value: 0.356025
         Iterations: 83
         Function evaluations: 180
         Gradient evaluations: 174
best_cost=0.012
X1 shape: torch.Size([100, 1])
Repetition  10:          Current function value: 0.356203
         Iterations: 49
         Function evaluations: 137
         Gradient evaluations: 131
best_cost=0.012
X1 shape: torch.Size([100, 1])
Repetition  11:          Current function value: 0.356307
         Iterations: 65
         Function evaluations: 162
         Gradient evaluations: 155
best_cost=0.012
X1 shape: torch.Size([100, 1])
Repetition  12:

/home/tu2/anaconda3/envs/deep-learning/lib/python3.13/site-
```

```
packages/scipy/optimize/_minimize.py:779: OptimizeWarning: Desired error not
necessarily achieved due to precision loss.
  res = _minimize_bfgs(fun, x0, args, jac, callback, **options)
/home/tu2/anaconda3/envs/deep-learning/lib/python3.13/site-
packages/scipy/optimize/_minimize.py:779: OptimizeWarning: Desired error not
necessarily achieved due to precision loss.
  res = _minimize_bfgs(fun, x0, args, jac, callback, **options)
/home/tu2/anaconda3/envs/deep-learning/lib/python3.13/site-
packages/scipy/optimize/_minimize.py:779: OptimizeWarning: Desired error not
necessarily achieved due to precision loss.
  res = _minimize_bfgs(fun, x0, args, jac, callback, **options)

         Current function value: 0.435275
         Iterations: 44
         Function evaluations: 147
         Gradient evaluations: 141
best_cost=0.012
X1 shape: torch.Size([100, 1])
Repetition  13:          Current function value: 0.356172
         Iterations: 30
         Function evaluations: 116
         Gradient evaluations: 111
best_cost=0.012
X1 shape: torch.Size([100, 1])
Repetition  14:          Current function value: 0.084118
         Iterations: 53
         Function evaluations: 141
         Gradient evaluations: 134
best_cost=0.012
X1 shape: torch.Size([100, 1])
Repetition  15:

/home/tu2/anaconda3/envs/deep-learning/lib/python3.13/site-
packages/scipy/optimize/_minimize.py:779: OptimizeWarning: Desired error not
necessarily achieved due to precision loss.
  res = _minimize_bfgs(fun, x0, args, jac, callback, **options)
/home/tu2/anaconda3/envs/deep-learning/lib/python3.13/site-
packages/scipy/optimize/_minimize.py:779: OptimizeWarning: Desired error not
necessarily achieved due to precision loss.
  res = _minimize_bfgs(fun, x0, args, jac, callback, **options)

         Current function value: 0.357745
         Iterations: 42
         Function evaluations: 143
         Gradient evaluations: 141
best_cost=0.012
X1 shape: torch.Size([100, 1])
Repetition  16:          Current function value: 0.309092
         Iterations: 52
```

```
         Function evaluations: 141
         Gradient evaluations: 136
best_cost=0.012
X1 shape: torch.Size([100, 1])
Repetition  17:

/home/tu2/anaconda3/envs/deep-learning/lib/python3.13/site-
packages/scipy/optimize/_minimize.py:779: OptimizeWarning: Desired error not
necessarily achieved due to precision loss.
  res = _minimize_bfgs(fun, x0, args, jac, callback, **options)
/home/tu2/anaconda3/envs/deep-learning/lib/python3.13/site-
packages/scipy/optimize/_minimize.py:779: OptimizeWarning: Desired error not
necessarily achieved due to precision loss.
  res = _minimize_bfgs(fun, x0, args, jac, callback, **options)

         Current function value: 0.355506
         Iterations: 197
         Function evaluations: 325
         Gradient evaluations: 320
best_cost=0.012
X1 shape: torch.Size([100, 1])
Repetition  18:          Current function value: 0.309409
         Iterations: 46
         Function evaluations: 131
         Gradient evaluations: 126
best_cost=0.012
X1 shape: torch.Size([100, 1])
Repetition  19:

/home/tu2/anaconda3/envs/deep-learning/lib/python3.13/site-
packages/scipy/optimize/_minimize.py:779: OptimizeWarning: Desired error not
necessarily achieved due to precision loss.
  res = _minimize_bfgs(fun, x0, args, jac, callback, **options)
/home/tu2/anaconda3/envs/deep-learning/lib/python3.13/site-
packages/scipy/optimize/_minimize.py:779: OptimizeWarning: Desired error not
necessarily achieved due to precision loss.
  res = _minimize_bfgs(fun, x0, args, jac, callback, **options)
/home/tu2/anaconda3/envs/deep-learning/lib/python3.13/site-
packages/scipy/optimize/_minimize.py:779: OptimizeWarning: Desired error not
necessarily achieved due to precision loss.
  res = _minimize_bfgs(fun, x0, args, jac, callback, **options)

         Current function value: 0.006044
         Iterations: 296
         Function evaluations: 399
         Gradient evaluations: 390
best_cost=0.006
X1 shape: torch.Size([100, 1])
Repetition  20:          Current function value: 0.353544
         Iterations: 75
```

```
         Function evaluations: 170
         Gradient evaluations: 159
best_cost=0.006
X1 shape: torch.Size([100, 1])
Repetition  21:          Current function value: 0.355939
         Iterations: 80
         Function evaluations: 180
         Gradient evaluations: 173
best_cost=0.006
X1 shape: torch.Size([100, 1])
Repetition  22:          Current function value: 0.262423
         Iterations: 59
         Function evaluations: 171
         Gradient evaluations: 165
best_cost=0.006
X1 shape: torch.Size([100, 1])
Repetition  23:          Current function value: 0.435266
         Iterations: 44
         Function evaluations: 142
         Gradient evaluations: 134
best_cost=0.006
X1 shape: torch.Size([100, 1])
Repetition  24:
```

```
/home/tu2/anaconda3/envs/deep-learning/lib/python3.13/site-
packages/scipy/optimize/_minimize.py:779: OptimizeWarning: Desired error not
necessarily achieved due to precision loss.
  res = _minimize_bfgs(fun, x0, args, jac, callback, **options)
/home/tu2/anaconda3/envs/deep-learning/lib/python3.13/site-
packages/scipy/optimize/_minimize.py:779: OptimizeWarning: Desired error not
necessarily achieved due to precision loss.
  res = _minimize_bfgs(fun, x0, args, jac, callback, **options)
/home/tu2/anaconda3/envs/deep-learning/lib/python3.13/site-
packages/scipy/optimize/_minimize.py:779: OptimizeWarning: Desired error not
necessarily achieved due to precision loss.
  res = _minimize_bfgs(fun, x0, args, jac, callback, **options)
/home/tu2/anaconda3/envs/deep-learning/lib/python3.13/site-
packages/scipy/optimize/_minimize.py:779: OptimizeWarning: Desired error not
necessarily achieved due to precision loss.
  res = _minimize_bfgs(fun, x0, args, jac, callback, **options)
```

```
         Current function value: 0.009657
         Iterations: 128
         Function evaluations: 250
         Gradient evaluations: 242
best_cost=0.006
X1 shape: torch.Size([100, 1])
Repetition  25:          Current function value: 0.011622
         Iterations: 98
```

```
         Function evaluations: 201
         Gradient evaluations: 196
best_cost=0.006
X1 shape: torch.Size([100, 1])
Repetition  26:

/home/tu2/anaconda3/envs/deep-learning/lib/python3.13/site-
packages/scipy/optimize/_minimize.py:779: OptimizeWarning: Desired error not
necessarily achieved due to precision loss.
  res = _minimize_bfgs(fun, x0, args, jac, callback, **options)
/home/tu2/anaconda3/envs/deep-learning/lib/python3.13/site-
packages/scipy/optimize/_minimize.py:779: OptimizeWarning: Desired error not
necessarily achieved due to precision loss.
  res = _minimize_bfgs(fun, x0, args, jac, callback, **options)
/home/tu2/anaconda3/envs/deep-learning/lib/python3.13/site-
packages/scipy/optimize/_minimize.py:779: OptimizeWarning: Desired error not
necessarily achieved due to precision loss.
  res = _minimize_bfgs(fun, x0, args, jac, callback, **options)

         Current function value: 0.030370
         Iterations: 128
         Function evaluations: 236
         Gradient evaluations: 226
best_cost=0.006
X1 shape: torch.Size([100, 1])
Repetition  27:          Current function value: 0.506213
         Iterations: 6
         Function evaluations: 83
         Gradient evaluations: 77
best_cost=0.006
X1 shape: torch.Size([100, 1])
Repetition  28:          Current function value: 0.356124
         Iterations: 79
         Function evaluations: 190
         Gradient evaluations: 186
best_cost=0.006
X1 shape: torch.Size([100, 1])
Repetition  29:

/home/tu2/anaconda3/envs/deep-learning/lib/python3.13/site-
packages/scipy/optimize/_minimize.py:779: OptimizeWarning: Desired error not
necessarily achieved due to precision loss.
  res = _minimize_bfgs(fun, x0, args, jac, callback, **options)
/home/tu2/anaconda3/envs/deep-learning/lib/python3.13/site-
packages/scipy/optimize/_minimize.py:779: OptimizeWarning: Desired error not
necessarily achieved due to precision loss.
  res = _minimize_bfgs(fun, x0, args, jac, callback, **options)

         Current function value: 0.355775
         Iterations: 34
```

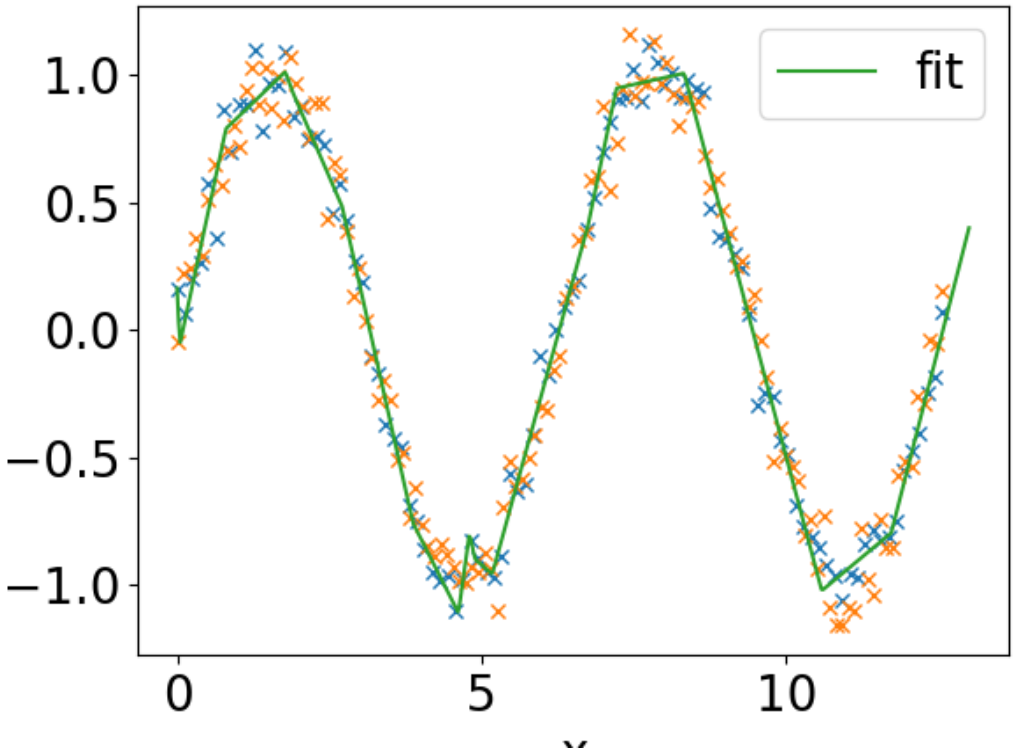
```
         Function evaluations: 117
         Gradient evaluations: 114
best_cost=0.006
X1 shape: torch.Size([100, 1])
Repetition  30:          Current function value: 0.098426
         Iterations: 108
         Function evaluations: 224
         Gradient evaluations: 216
best_cost=0.006
X1 shape: torch.Size([100, 1])
Repetition  31:
```

```
/home/tu2/anaconda3/envs/deep-learning/lib/python3.13/site-
packages/scipy/optimize/_minimize.py:779: OptimizeWarning: Desired error not
necessarily achieved due to precision loss.
  res = _minimize_bfgs(fun, x0, args, jac, callback, **options)
/home/tu2/anaconda3/envs/deep-learning/lib/python3.13/site-
packages/scipy/optimize/_minimize.py:779: OptimizeWarning: Desired error not
necessarily achieved due to precision loss.
  res = _minimize_bfgs(fun, x0, args, jac, callback, **options)
/home/tu2/anaconda3/envs/deep-learning/lib/python3.13/site-
packages/scipy/optimize/_minimize.py:779: OptimizeWarning: Desired error not
necessarily achieved due to precision loss.
  res = _minimize_bfgs(fun, x0, args, jac, callback, **options)
```

```
         Current function value: 0.355976
         Iterations: 66
         Function evaluations: 181
         Gradient evaluations: 176
best_cost=0.006
X1 shape: torch.Size([100, 1])
Repetition  32: Optimization terminated successfully.
         Current function value: 0.506238
         Iterations: 3
         Function evaluations: 5
         Gradient evaluations: 5
best_cost=0.006
X1 shape: torch.Size([100, 1])
Repetition  33:          Current function value: 0.355819
         Iterations: 44
         Function evaluations: 134
         Gradient evaluations: 129
best_cost=0.006
X1 shape: torch.Size([100, 1])
Repetition  34: Optimization terminated successfully.
         Current function value: 0.506238
         Iterations: 6
         Function evaluations: 7
         Gradient evaluations: 7
```

```
best_cost=0.006
X1 shape: torch.Size([100, 1])
Repetition  35:          Current function value: 0.355245
         Iterations: 84
         Function evaluations: 184
         Gradient evaluations: 178
best_cost=0.006
X1 shape: torch.Size([100, 1])
Repetition  36:

/home/tu2/anaconda3/envs/deep-learning/lib/python3.13/site-
packages/scipy/optimize/_minimize.py:779: OptimizeWarning: Desired error not
necessarily achieved due to precision loss.
  res = _minimize_bfgs(fun, x0, args, jac, callback, **options)
/home/tu2/anaconda3/envs/deep-learning/lib/python3.13/site-
packages/scipy/optimize/_minimize.py:779: OptimizeWarning: Desired error not
necessarily achieved due to precision loss.
  res = _minimize_bfgs(fun, x0, args, jac, callback, **options)
/home/tu2/anaconda3/envs/deep-learning/lib/python3.13/site-
packages/scipy/optimize/_minimize.py:779: OptimizeWarning: Desired error not
necessarily achieved due to precision loss.
  res = _minimize_bfgs(fun, x0, args, jac, callback, **options)

         Current function value: 0.356222
         Iterations: 53
         Function evaluations: 154
         Gradient evaluations: 148
best_cost=0.006
X1 shape: torch.Size([100, 1])
Repetition  37:          Current function value: 0.356919
         Iterations: 56
         Function evaluations: 129
         Gradient evaluations: 123
best_cost=0.006
X1 shape: torch.Size([100, 1])
Repetition  38: Optimization terminated successfully.
         Current function value: 0.437668
         Iterations: 15
         Function evaluations: 18
         Gradient evaluations: 18
best_cost=0.006
X1 shape: torch.Size([100, 1])
Repetition  39: Optimization terminated successfully.
         Current function value: 0.506238
         Iterations: 4
         Function evaluations: 5
         Gradient evaluations: 5
best_cost=0.006
X1 shape: torch.Size([100, 1])
```

```
Repetition  40:          Current function value: 0.356435
         Iterations: 61
         Function evaluations: 147
         Gradient evaluations: 141
best_cost=0.006
X1 shape: torch.Size([100, 1])
Repetition  41:

/home/tu2/anaconda3/envs/deep-learning/lib/python3.13/site-
packages/scipy/optimize/_minimize.py:779: OptimizeWarning: Desired error not
necessarily achieved due to precision loss.
  res = _minimize_bfgs(fun, x0, args, jac, callback, **options)
/home/tu2/anaconda3/envs/deep-learning/lib/python3.13/site-
packages/scipy/optimize/_minimize.py:779: OptimizeWarning: Desired error not
necessarily achieved due to precision loss.
  res = _minimize_bfgs(fun, x0, args, jac, callback, **options)

         Current function value: 0.084814
         Iterations: 66
         Function evaluations: 148
         Gradient evaluations: 142
best_cost=0.006
X1 shape: torch.Size([100, 1])
Repetition  42:          Current function value: 0.268998
         Iterations: 83
         Function evaluations: 186
         Gradient evaluations: 182
best_cost=0.006
X1 shape: torch.Size([100, 1])
Repetition  43: Optimization terminated successfully.
         Current function value: 0.506238
         Iterations: 10
         Function evaluations: 15
         Gradient evaluations: 15
best_cost=0.006
X1 shape: torch.Size([100, 1])
Repetition  44:

/home/tu2/anaconda3/envs/deep-learning/lib/python3.13/site-
packages/scipy/optimize/_minimize.py:779: OptimizeWarning: Desired error not
necessarily achieved due to precision loss.
  res = _minimize_bfgs(fun, x0, args, jac, callback, **options)
/home/tu2/anaconda3/envs/deep-learning/lib/python3.13/site-
packages/scipy/optimize/_minimize.py:779: OptimizeWarning: Desired error not
necessarily achieved due to precision loss.
  res = _minimize_bfgs(fun, x0, args, jac, callback, **options)

         Current function value: 0.355569
         Iterations: 74
         Function evaluations: 164
```

```
         Gradient evaluations: 157
best_cost=0.006
X1 shape: torch.Size([100, 1])
Repetition  45:          Current function value: 0.355443
         Iterations: 149
         Function evaluations: 270
         Gradient evaluations: 265
best_cost=0.006
X1 shape: torch.Size([100, 1])
Repetition  46:

/home/tu2/anaconda3/envs/deep-learning/lib/python3.13/site-
packages/scipy/optimize/_minimize.py:779: OptimizeWarning: Desired error not
necessarily achieved due to precision loss.
  res = _minimize_bfgs(fun, x0, args, jac, callback, **options)
/home/tu2/anaconda3/envs/deep-learning/lib/python3.13/site-
packages/scipy/optimize/_minimize.py:779: OptimizeWarning: Desired error not
necessarily achieved due to precision loss.
  res = _minimize_bfgs(fun, x0, args, jac, callback, **options)
/home/tu2/anaconda3/envs/deep-learning/lib/python3.13/site-
packages/scipy/optimize/_minimize.py:779: OptimizeWarning: Desired error not
necessarily achieved due to precision loss.
  res = _minimize_bfgs(fun, x0, args, jac, callback, **options)

         Current function value: 0.268301
         Iterations: 115
         Function evaluations: 225
         Gradient evaluations: 218
best_cost=0.006
X1 shape: torch.Size([100, 1])
Repetition  47:          Current function value: 0.083666
         Iterations: 113
         Function evaluations: 215
         Gradient evaluations: 205
best_cost=0.006
X1 shape: torch.Size([100, 1])
Repetition  48:          Current function value: 0.435218
         Iterations: 32
         Function evaluations: 119
         Gradient evaluations: 114
best_cost=0.006
X1 shape: torch.Size([100, 1])
Repetition  49: Optimization terminated successfully.
         Current function value: 0.372736
         Iterations: 27
         Function evaluations: 34
         Gradient evaluations: 34
best_cost=0.006
Training error: 0.00604387978091836
```

```
Test error    : 0.012707690708339214
```